IE540
L4. Monte Carlo Method

2021 Spring

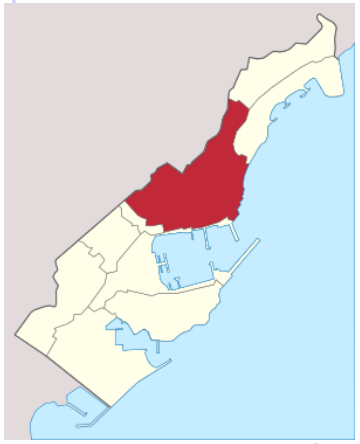
KAIST ISE
Hayong Shin

Overview

- Model is known → DP (in Lecture 3)
  - Evaluation (prediction) : Iterative policy evaluation
  - Control (finding optimal solution)
    - PI → GPI → VI
    - Async. DP
- Model is unknown → RL (by use of interaction samples)
  - Value based RL
    - MC method : Evaluation, Control (in Lecture 4)
    - TD method : Q, SARSA (in Lecture 5)
    - n-step bootstrapping (in Lecture 6)
    - value function approximation (in Lecture 7)
    - NN, DQN (in Lecture 8)
  - Policy based RL (Lecture 9~11)
    - Derivative free methods
    - Policy gradient
- Preliminary for RL
  - Concept of MC
  - Stochastic approximation

Monte Carlo method

- Monaco : Queen Grace Kelly, F1 race track, football team, and
- A part of Monaco (one of the four quarters) : Monte Carlo



https://en.wikipedia.org/wiki/File:Monte_Carlo_in_Monaco.svg



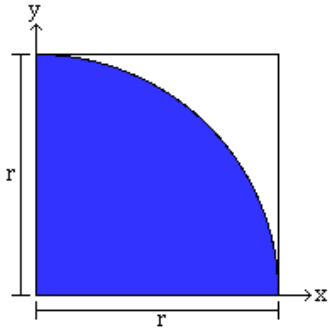
https://commons.wikimedia.org/wiki/File:Monaco_Monte_Carlo_1.jpg

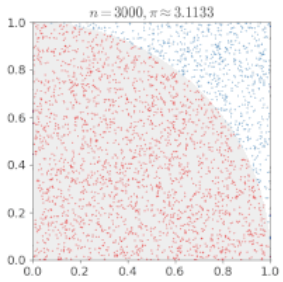


https://www.pokerstars.com/en/news/the-oldest-casinos-in-europe-and-las-vegas/61988/

- MC method
  - a.k.a. MC simulation, MC integration
  - Computing integration ($\approx$ expectation) using samples







$$\pi = 4 \frac{\#\text{ darts hitting shaded area}}{\#\text{ darts hitting inside square}}$$

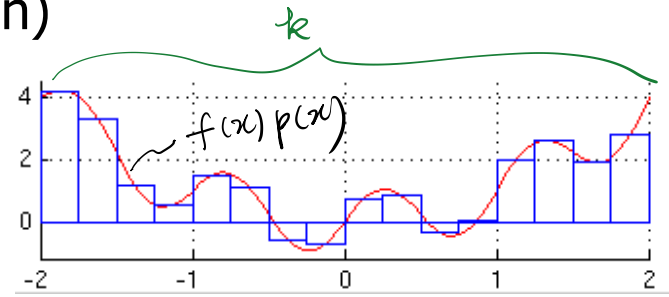
https://pt.slideshare.net/rpiryani/monte-carlo-simulation-44005705/5

History of Monte Carlo Method

- In 1940's, physicists working on Manhattan project
  - in Los Alamos National Lab.
  - in the quest of the distance that neutrons would likely travel through various materials
  - Stanislaw Ulam with John von Neumann suggested to solve the problem by computer experiment using chance
  - The name "Monte Carlo" was given by Nicholas Metropolis.

- Since then, it found many applications in science and engineering
  - High energy physics
  - Risk assessment
  - Financial derivative pricing
  - Machine learning incl. RL
  - …

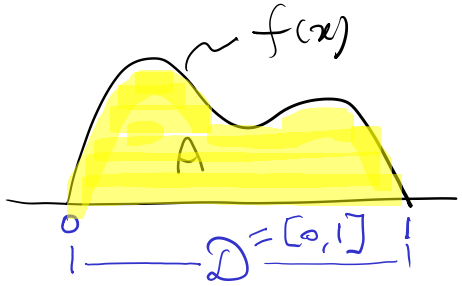
MC method (aka MC simulation, MC integration)



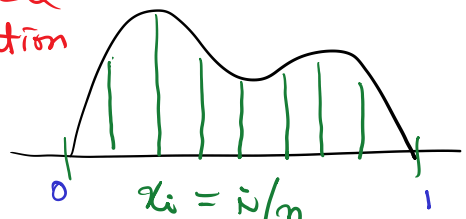
- Computing expectation $\mu = \mathbb{E}_{X \in \mathcal{D} \sim p(x)}[f(X)]$
  - discrete $\mu = \sum_{x \in \mathcal{D}} f(x)p(x)$
  - continuous $\mu = \int_{\mathcal{D}} f(x)p(x)dx = \int_{\mathcal{D}} f(x)dP(x)$
  - no closed form solution → numerical integration (or summation)
    - unapplicable in high dim.
      - $d = \dim(\mathcal{D})$, $k = \#$ of intervals in an axis → $N = k^d$ evaluations
        - » $k = 2,\ d = 100 \rightarrow N = 2^{100} \cong 10^{30}$
      - What if we can afford only 1M(10^6) evaluations of $f(x)p(x)$?
    - unapplicable if $p(x)$ is unknown (but, samples from $p(x)$ are available)
- MC method : Computing expectation by
  - $x^{(1)}, \ldots, x^{(n)} \sim p(x)$: i.i.d. samples from $p(x)$
  - $\mu_n = \frac{1}{n}\sum_i f\left(x^{(i)}\right)$: unbiased estimator of μ
    - $\mathbb{E}[\mu_n] = \mathbb{E}\left[\frac{1}{n}\sum_i f\left(x^{(i)}\right)\right] = \frac{1}{n}\sum_i \mathbb{E}\left[f\left(x^{(i)}\right)\right] = \frac{1}{n}\sum_i \mu = \mu$
    - $\mathrm{Var}[\mu_n] = \mathrm{Var}\left[\frac{1}{n}\sum_i f\left(x^{(i)}\right)\right] = \frac{1}{n^2}\sum_i \mathrm{Var}\left[f\left(x^{(i)}\right)\right] = \frac{\sigma^2}{n}$, where $\sigma^2 = \mathrm{Var}_{X \sim p(x)}[f(X)]$
    - $\mu_n \rightarrow \mu$ as $n \rightarrow \infty$
  - applicable when d is high or $p(x)$ is unknown

MC Integration : Computing integral using random samples

- Computing an integral ≡ Computing an expectation
  - $A = \int_0^1 f(x)dx = \int_0^1 f(x) \cdot 1 dx = \mathbb{E}$ [ ] $[f(X)]$
  - MC Integration
    - $x^{(1)}, \dots, x^{(n)} \sim U(0,1)$: i.i.d. samples from $U(0,1)$
    - $A_n = \frac{1}{n}\sum_i f\left(x^{(i)}\right)$: unbiased estimator of A



Numerical integration



→ regular split of $\mathcal{D}$

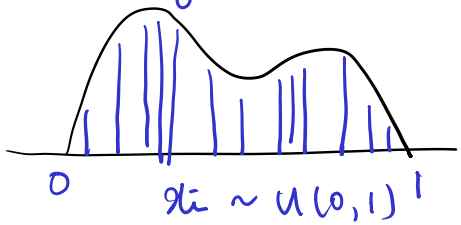
- In general, $A = \int_{\mathcal{D}} f(x)dx$
  - Consider a random variable $X \in \mathcal{D} \sim p(x)$
    - $p(x)$: any convenient distribution with domain $\mathcal{D}$

$$A = \int_{\mathcal{D}} f(x)dx = \int_{\mathcal{D}} \frac{f(x)}{p(x)} p(x)dx = \mathbb{E}$$ [ ] [ ]

    - MC estimator of A

$$A_n = \frac{1}{n}\sum_{i=1}^{n}$$ [ ] where $x^{(i)} \sim$ [ ]

MC integration



↳ random sample

Variance Reduction

- How can we reduce the variance of an estimator
  - Increase the number of simulation
    - Standard error of $\mu_n = \sqrt{\mathrm{Var}[\mu_n]} = \frac{\sigma}{\sqrt{n}} : O\left(\frac{1}{\sqrt{n}}\right)$
- More evenly distributed sequence
  - Stratification or Latin Hypercube Sampling
  - cf. Quasi-MC : use deterministic low-discrepancy sequence
    - Sobol sequence

- Variance reduction techniques
  - Antithetic variable
  - Control variate
  - Importance sampling
  - Common random number

- Goal : $\mu = \mathbb{E}_{X\sim p(x)}[f(X)]$
- Samples are from different dist. $q(x)$
  - $x^{(1)}, \dots, x^{(n)} \sim q(x)$
- Rewrite μ
  - $\mu = \int_{\mathcal{D}} f(x)p(x)dx =$

    $\int_{\mathcal{D}} f(x)\,\frac{p(x)}{q(x)}\,q(x)dx = \mathbb{E}_{X\sim}\square\left[f(X)\square\right]$
- Importance sampling estimator
  - $\mu_n = \frac{1}{n}\sum_{i=1}^{n} f(x^{(i)})\frac{p(x^{(i)})}{q(x^{(i)})}$ — importance weight

Incremental average

- Given data $x^{(1)}, \ldots, x^{(n)}$ → find average
- Simple arithmetic average
  - $\mu_n = \frac{1}{n}\left(x^{(1)} + x^{(2)} + \cdots + x^{(n)}\right)$
- Incremental arithmetic average
  - Assume that data x_i is given one by one
  - $\mu_n = \frac{1}{n}\left(x^{(1)} + \cdots + x^{(n)}\right) = \frac{1}{n}\left[\left(x^{(1)} + \cdots + x^{(n-1)}\right) + x^{(n)}\right] = \frac{1}{n}\left[(n-1)\mu_{n-1} + x^{(n)}\right]$
  - $\mu_n = \frac{n-1}{n}\mu_{n-1} + \frac{1}{n}x^{(n)} = \mu_{n-1} + \frac{1}{n}$ [ ]
    - $\mu_0 = 0$: initial estimate (it has no effect in this case)
    - μ_{n-1} : previous estimate, x_n : new data
    - $\left(x^{(n)} - \mu_{n-1}\right)$: surprise of new data
- TAQ. Incremental weighted average?
  - $\mu_n = \sum_{i=1}^{n} w_i x^{(i)} \Big/ \sum_{i=1}^{n} w_i$ w_i : weight of data $x^{(i)}$

$$\mu_n = \frac{\sum_{i=1}^{n} w_i x^{(i)}}{\sum_{i=1}^{n} w_i} = \frac{\sum_{i=1}^{n} w_i x^{(i)}}{C_n} = \frac{\sum_{i=1}^{n-1} w_i x^{(i)} + w_n x^{(n)}}{C_n}$$

$$= \frac{C_{n-1}}{C_n}\mu_{n-1} + \frac{w_n}{C_n} x^{(n)} = \mu_{n-1} + \frac{w_n}{C_n}\left(x^{(n)} - \mu_{n-1}\right)$$

$$C_n = \sum_{i=1}^{n} w_n$$

Generalized incremental average

- Incremental arithmetic average: $\mu_n = \mu_{n-1} + \frac{1}{n}(x^{(n)} - \mu_{n-1})$
- Generalized incremental average: $\mu_n = \mu_{n-1} + \alpha_n(x^{(n)} - \mu_{n-1})$
  - α_n : step size, learning rate
    - μ_n learns from the surprise of new data $(x^{(n)} - \mu_{n-1})$ at the rate of α_n
    - meaningful range of α_n :
  - $\alpha_n = \alpha$ (constant) → EWMA(exponentially weighted moving average), exponential smoothing
    - $\mu_n = \alpha x^{(n)} + (1-\alpha)\mu_{n-1} = \alpha x^{(n)} + (1-\alpha)\left[\alpha x^{(n-1)} + (1-\alpha)\mu_{n-2}\right]$
      $= \alpha\left\{x^{(n)} + (1-\alpha)x^{(n-1)} + (1-\alpha)^2 x^{(n-2)} + \cdots + (1-\alpha)^{n-1}x^{(1)}\right\} + (1-\alpha)^n \mu_0$
    - eg: $\alpha = 0.1, n = 3 \rightarrow \mu_3 = 0.1x^{(3)} + 0.09x^{(2)} + 0.081x^{(1)} + 0.729\mu_0$
      → bigger weights on recent data
    - TAQ. When $\alpha_n = \alpha$, $\mathrm{Var}[\mu_n] \rightarrow 0$ as $n \rightarrow \infty$?
  - $\alpha_n = \frac{1}{n}$ → simple arithmetic average
  - $\alpha_n = \frac{1}{n_0+n}$ → μ_0 has the weight of samples
    - $\mu_1 = \frac{n_0}{n_0+1}\mu_0 + \frac{1}{n_0+1}x^{(1)}$, $\mu_2 = \frac{n_0}{n_0+2}\mu_0 + \frac{1}{n_0+2}x^{(1)} + \frac{1}{n_0+2}x^{(2)}$, ...

Robbins-Monro condition on step size α_n

- Q: Does μ_n converge to $\mu = \mathbb{E}[x^{(i)}]$?
- Convergence of generalized incremental average
  - from any μ_0, (regardless of initial estimate μ_0)
  - $\mu_n = \mu_{n-1} + \alpha_n(x^{(n)} - \mu_{n-1})$ converge to $\mu = \mathbb{E}[x^{(i)}]$
  - if step size α_n satisfies the following 2 conditions (R-M cond.)

    (1) $\sum_{n=1}^{\infty} \alpha_n^2 < \infty$
    - this implies that [ ] as $n \to \infty$
    - step size should decrease fast enough

    (2) $\sum_{n=1}^{\infty} \alpha_n = \infty$
    - but it should not decrease too fast
    - so that μ_n can reach anywhere
- examples of α_n
  - $\alpha_n \geq \alpha$ → (1) is violated → does not converge
  - $\alpha_n = e^{-n}$ → (2) is violated → premature convergence
  - $\alpha_n = \frac{1}{n^p}$ → (1) & (2) are satisfied if [ ]

Convergence of $\sum_{n=1}^{\infty} \frac{1}{n^p}$

$$\sum_{n=1}^{\infty} \frac{1}{n} = \left\{\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}+\cdots\right\} = \frac{1}{1}+\frac{1}{2}+\left(\frac{1}{3}+\frac{1}{4}\right)+\left(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}\right)+\cdots$$
$$> \frac{1}{1}+\frac{1}{2}+\left(\frac{1}{4}+\frac{1}{4}\right)+\left(\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}\right)+\cdots = \frac{1}{1}+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\cdots = \infty$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < 1+\sum_{n=2}^{\infty} \frac{1}{n(n-1)} = 1+\sum_{n=2}^{\infty}\left\{\frac{1}{n-1}-\frac{1}{n}\right\} = 1+\left\{\frac{1}{1}-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\cdots\right\} = 2$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \approx \int_1^{\infty} \frac{1}{x^p}dx = \begin{cases} \left.\frac{x^{1-p}}{1-p}\right|_1^{\infty} = \infty, & \text{if } p<1 \\ \left.\log x\right|_1^{\infty} = \infty, & \text{if } p=1 \\ \left.\frac{-1}{(p-1)x^{p-1}}\right|_1^{\infty} = \frac{1}{p-1}, & \text{if } p>1 \end{cases} \rightarrow \begin{cases} = \infty \text{ (diverges)} & \text{if } p \le 1 \\ < \infty \text{ (converges)} & \text{if } p > 1 \end{cases}$$

$$\alpha_n = \frac{1}{n^p} \rightarrow \left\{\begin{array}{l} (1)\ \sum_{n=1}^{\infty} \alpha_n^2 < \infty \rightarrow 2p > 1 \\ (2)\ \sum_{n=1}^{\infty} \alpha_n = \infty \rightarrow p \le 1 \end{array}\right\} \rightarrow \frac{1}{2} < p \le 1$$

THEOREM 9—The Integral Test Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $x \geq N$ (N a positive integer). Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x)\,dx$ both converge or both diverge.

Caution
The series and integral need not have the same value in the convergent case. You will see in Example 6 that

$$\sum_{n=1}^{\infty} (1/n^2) \neq \int_1^{\infty} (1/x^2)\,dx = 1.$$

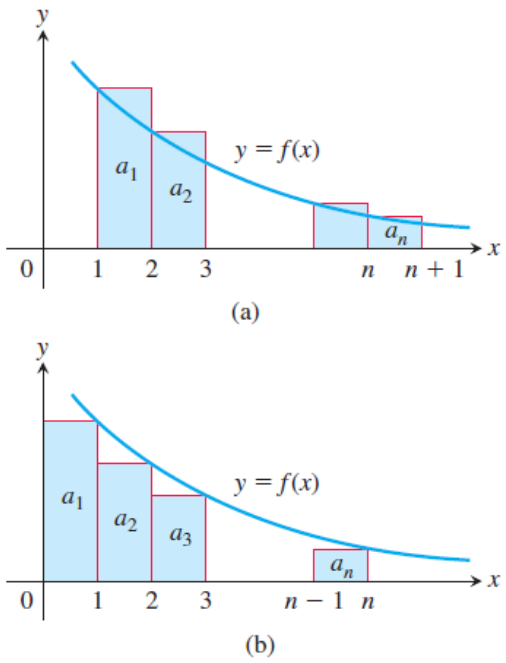


FIGURE 10.11 Subject to the conditions of the Integral Test, the series $\sum_{n=1}^{\infty} a_n$ and the integral $\int_1^{\infty} f(x)\,dx$ both converge or both diverge.

Proof We establish the test for the case $N = 1$. The proof for general N is similar.

We start with the assumption that f is a decreasing function with $f(n) = a_n$ for every n. This leads us to observe that the rectangles in Figure 10.11a, which have areas $a_1, a_2, \ldots, a_n$, collectively enclose more area than that under the curve $y = f(x)$ from $x = 1$ to $x = n + 1$. That is,

$$\int_1^{n+1} f(x)\,dx \leq a_1 + a_2 + \cdots + a_n.$$

In Figure 10.11b the rectangles have been faced to the left instead of to the right. If we momentarily disregard the first rectangle of area a_1, we see that

$$a_2 + a_3 + \cdots + a_n \leq \int_1^{n} f(x)\,dx.$$

If we include a_1, we have

$$a_1 + a_2 + \cdots + a_n \leq a_1 + \int_1^{n} f(x)\,dx.$$

Combining these results gives

$$\int_1^{n+1} f(x)\,dx \leq a_1 + a_2 + \cdots + a_n \leq a_1 + \int_1^{n} f(x)\,dx.$$

These inequalities hold for each n, and continue to hold as $n \to \infty$.

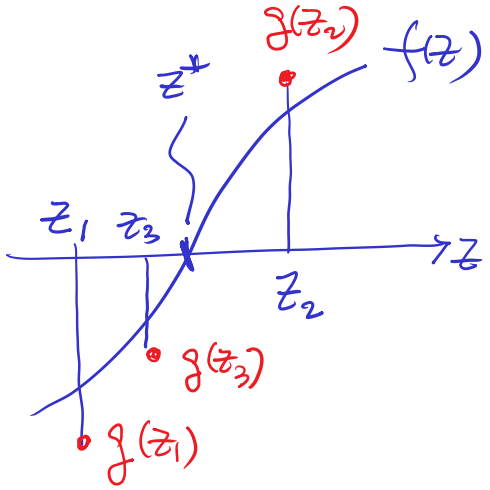
If $\int_1^{\infty} f(x)\,dx$ is finite, the right-hand inequality shows that $\sum a_n$ is finite. If $\int_1^{\infty} f(x)\,dx$ is infinite, the left-hand inequality shows that $\sum a_n$ is infinite. Hence the series and the integral are both finite or both infinite. ■

Thomas' Calculus, p.597

Stochastic approximation

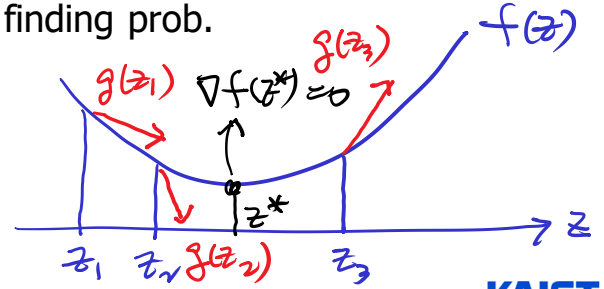
- Hubbert Robbins, Sutton Monro, "A stochastic approximation method", Annals of Math. Stat, 22(3), 1951
- Problem : root finding with noise observation
  - find x s.t. $f(z) = 0$
    - assume $f(z)$ is an increasing function
    - only noisy observation $g(z)$ is available $\left(\mathbb{E}[g(z)] = f(z)\right)$
      - eg. additive noise $g(z) = f(z) + \epsilon$, where $\mathbb{E}[\epsilon] = 0$
  - stochastic approximation algo. (R-M algo.)
    - $z_{n+1} = z_n - \alpha_n g(z_n)$
    - if α_n satisfies the following 2 conditions (R-M cond.)
      - $\sum_{n=1}^{\infty} \alpha_n^2 < \infty$
      - $\sum_{n=1}^{\infty} \alpha_n = \infty$

    → z_n converges to z^* where $f(z^*) = 0$



- Problem : Local optimization (minimization) with stochastic gradient
  - find x s.t. $\nabla f(z) = 0$
    - let $g(z) = \nabla f(z) + \epsilon$, where $\mathbb{E}[\epsilon] = 0$ → Same as root finding prob.
    - $z_{n+1} = z_n - \alpha_n g(z_n)$

    → z_n converges to z^* s.t. $\nabla f(z^*) = 0$
    if R-M conditions are met.



Stochastic approximation

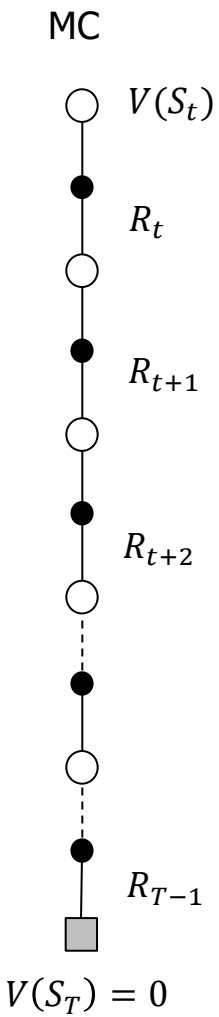
- Applying R-M for incremental averaging
  - Given data : $x^{(1)}, \ldots, x^{(n)}, \ldots \sim \text{iid } p(x)$
  - Problem : to find $\mathbb{E}_{X\sim p(x)}[X]$ using $x^{(i)}$ incrementally
  - Consider $f(z) =$
  - Let $g(z) = z - x^{(n)}$: nth evaluation of $f(z)$
    - $\mathbb{E}_{x^{(n)}\sim p(x)}[g(z)] = f(z)$
  - By applying R-M stochastic approximation, we get
    - $z_{n+1} = z_n - \alpha_n g(z_n) = z_n - \alpha_n\left(z_n - x^{(n)}\right) = z_n + \alpha_n\left(x^{(n)} - z_n\right)$
    - let $\mu_n = z_{n+1}$ → $\mu_n = \mu_{n-1} + \alpha_n\left(x^{(n)} - \mu_{n-1}\right)$
    - μ_n converge to $\mu = \mathbb{E}_{X\sim p(x)}[X]$ if

      (1) $\sum_{n=1}^{\infty} \alpha_n^2 < \infty$

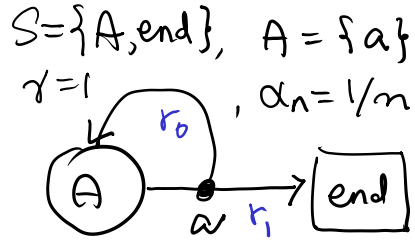
      (2) $\sum_{n=1}^{\infty} \alpha_n = \infty$

Monte Carlo Policy Evaluation (Prediction)

- No model ($p^a_{s,s'}$ and $R(s,a)$) → can't use DP
- Sample trajectory(= sample episode, sample path)
  - $\tau = (S_0, A_0, R_0, S_1, A_1, R_1, \dots, S_T)$
    - partial trajectory $\tau_t = (S_t, A_t, R_t, S_{t+1}, A_{t+1}, R_{t+1}, \dots, S_T)$
  - $\tau^{(i)} \sim \pi$: i^{th} sample trajectory
    - $\tau_t^{(i)}$: i^{th} partial trajectory from time t
- Return : total discounted reward
  - $G_t = R_t + \gamma R_{t+1} + \cdots + \gamma^{T-t-1} R_{T-1}$
  - $G_t^{(i)}$: return of $\tau_t^{(i)}$
- MC PE (Monte Carlo Policy Evaluation)
  - $V^\pi(s) = \mathbb{E}_\pi[G_t | S_t = s]$
  - $V(s) \approx \frac{1}{n}\sum_{i=1}^{n} G_t^{(i)}$?
    - How to ensure $S_t = s$ for each $\tau_t^{(i)}$?
    - Instead, for each S_t of $\tau^{(i)}$, Update $V(S_t)$ incrementally.



Monte Carlo Policy Evaluation



$\tau = (A, a, r_0, A, a, r_1, end)$

➢ **Algorithm MC PE (every visit)**

1. initialize $V(s) = 0, N(s) = 0$ for $\forall s$
2. for $i = 1..n$ (n : # of episodes) (or until convergence)
   - get an episode $\tau = (S_0, A_0, R_0, S_1, A_1, R_1, \dots, S_T)$
   - for $t = T-1..0$ (compute G_t backward from $G_T = 0$)
     - $G_t = R_t + \gamma G_{t+1}$
     - $V(S_t) \leftarrow V(S_t) + \alpha_t\big(G_t - V(S_t)\big)$

// first visit MC PE
if $S_t = S_{t'}$ for some $t' < t$
    // visited S_t before, in τ
    // skip it.
    continue

Every visit N(A)

V(A)

=

=

- $V(s) \to V^{\pi}(s)$ as $N(s) \to \infty$ if α_t satisfies R-M cond.
  - $\alpha_t = \frac{1}{N(S_t)}$: simple average
    - need to track $N(s)$ for each state s
  - $\alpha_t = \alpha$: exponential smoothing
    - never converge under certain limit
    - useful for non-stationary env.
      - track a running mean of recent env.

First visit

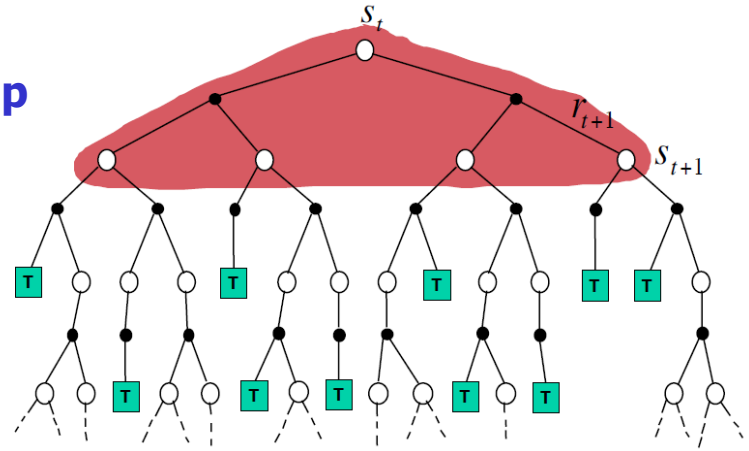
V(A)

=

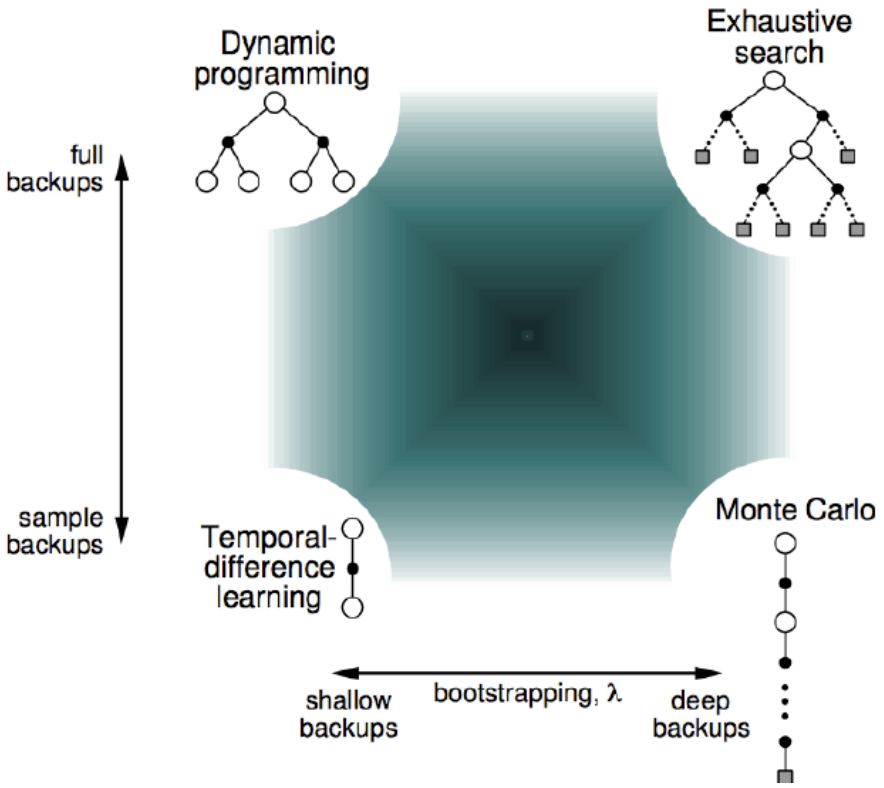
Monte Carlo Policy Evaluation

- Every visit or First visit? → FV seems more intuitive, but both are fine
  - Do FV and EV converge to the same value? not exactly.
    - FV and EV converge to $V^{\pi}(s)$ with different model parameters
    - FV → replacing trace TD(1), EV → accumulating trace TD(1)
    - See Singh and Sutton, 1996
- Will MC PE cover $V(s)$ for every s?
  - No. $V(s)$ will be updated only for state s visited by the policy π
- MC PE for action-value function $Q^{\pi}(s,a)$? → will be covered in MC control
  - update $N(S_t, A_t)$ and $Q(S_t, A_t)$ in the same manner
  - $Q(s,a)$ will be updated for visited state-action (s,a) by the policy π
- Bias? → G_t is an unbiased estimator of $V^{\pi}(S_t)$
  - $V^{\pi}(s) = \mathbb{E}_{\pi}[G_t | S_t = s]$
- Variance of MC PE : $\mathrm{Var}\left[G_t^{(i)}\right]$ is high.
  - $\mathrm{Var}\left[\frac{1}{n}\sum_{i=1}^{n} G_t^{(i)}\right] = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}\left[G_t^{(i)}\right]$,
  - $\mathrm{Var}\left[G_t^{(i)}\right] = \mathrm{Var}[R_t + \gamma R_{t+1} + \cdots + \gamma^{T-t} R_T] \gg \mathrm{Var}[R_t]$

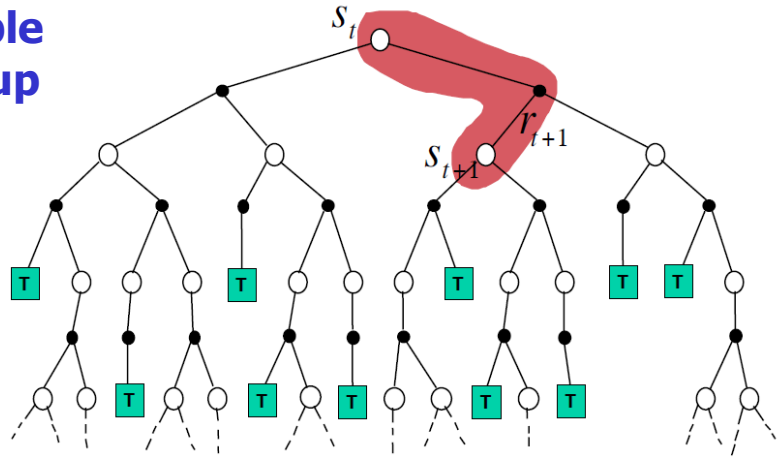
Full backup



DP Backup : $V(S_t) = \mathbb{E}_\pi[R_t + \gamma V(S_{t+1})]$

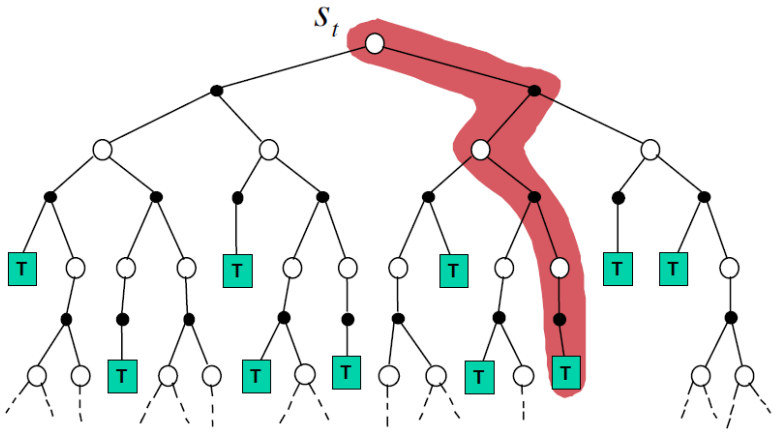


Sample backup



TD Backup

$$V(S_t) \leftarrow V(S_t) + \alpha_t\big(R_t + \gamma V(S_{t+1}) - V(S_t)\big)$$



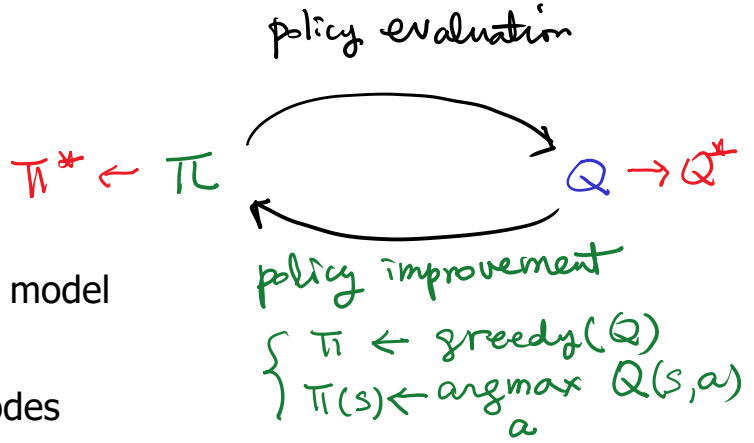
MC Backup

$$V(S_t) \leftarrow V(S_t) + \alpha_t\big(G_t - V(S_t)\big)$$

https://medium.com/intro-to-artificial-intelligence/q-learning-a-value-based-reinforcement-learning-algorithm-272706d835cf

Monte Carlo Control

- GPI
  - policy evaluation → [redacted]
  - policy improvement → greedy
- value function : V or Q
  - for control, Q(action value) is better
    - in DP, Q can be directly obtained from V using model
      - $Q^{\pi}(s,a) = R(s,a) + \gamma \mathbb{E}_{s' \sim p^{a}_{s,s'}}[V^{\pi}(s')]$
    - w/o model, learning $Q(s,a)$ directly from episodes



➢ **Algorithm MC Control w/ ES (Exploring Starts)**

1. initialize $Q(s,a) = 0, N(s,a) = 0$ for $\forall(s,a)$
2. for $k = 1..n$ (n : # of episodes) (or until convergence)
   - get an episode $\tau = (S_0, A_0, R_0, S_1, A_1, R_1, \ldots, S_T)$
     - from randomly chosen (S_0, A_0), with P$[(S_0, A_0)]$>0 for any (S_0, A_0) (exploring start)
     - then, following greedy policy π
   - for $t = T-1..0$ (compute G_t backward from $G_T = 0$)
     - $G_t = R_t + \gamma G_{t+1}$
     - if $(S_t, A_t) = (S_{t'}, A_{t'})$ for some $t' < t$, continue (first visit MC)
     - $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha_t\big(G_t - Q(S_t, A_t)\big)$
       - usually use $\alpha_t = \frac{1}{N(S_t, A_t)}$ which requires to track $N(S_t, A_t)$
     - $\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$ (policy improvement)

For control, we need [redacted]
Selecting (S_0, A_0) is the only exploration here, but usually we cannot choose S_0.

GLIE policy

- GLIE policy (= soft policy) → MC control w/ GLIE policy converges to optimal $Q^*(s,a)$
  - GLIE : Greedy in the Limit with Infinite Exploration
    - GL : the policy converges on a greedy policy
      - $\pi_k \to \text{greedy}(Q_k)$ as $k \to \infty$
        - » π_k: policy, Q_k: action value function at iteration k
        - » i.e. $\pi_k(a|s) \to 1$ for $a = \arg\max_{a'} Q_k(s,a')$ as $k \to \infty$
    - IE : all state-action pairs are explored infinitely many times
      - $N_k(s,a) \to \infty$ as $k \to \infty$
- ε-greedy(Q) policy : random action wp ε, greedy action wp 1- ε
  - $\pi(a|s) = \begin{cases} \epsilon/|A(s)|, & \text{if } a \neq a^*(s) \\ 1-\epsilon+\epsilon/|A(s)|, & \text{if } a = a^*(s) \end{cases}$, where $a^*(s) = \arg\max_{a'} Q(s,a')$
  - conditions for GLIE
    - greedy as $k \to \infty$ ➔ $\lim_{k\to\infty} \epsilon_k = 0$
    - $N_k(s,a) \propto \sum_{n=1}^{k} \epsilon_n$ ➔ $\sum_{k=1}^{\infty} \epsilon_k = \infty$

Examples

$\epsilon_k = e^{-k}$ ➔ $\sum_{k=1}^{\infty} \epsilon_k$

$\epsilon_k = \frac{1}{k^p}$ ➔ GLIE if

On-policy MC control

- Control : finding optimal value/policy
  - on-policy control : behavior policy = target policy
    - behavior policy : policy to generate episodes
    - target policy : policy we want obtain at the end of learning
  - off-policy control : behavior policy ≠ target policy

➢ **Algorithm On-policy MC Control**

1. initialize $Q(s,a) = 0, N(s,a) = 0$ for $\forall(s,a)$
2. for $k = 1..n$ (n : # of episodes) (or until convergence)
   - get an episode $\tau = (S_0, A_0, R_0, S_1, A_1, R_1, \dots, S_T)$
     - using $\pi = \epsilon - \text{greedy}(Q)$
   - for $t = T-1..0$ (compute G_t backward from $G_T = 0$)
     - $G_t = R_t + \gamma G_{t+1}$
     - if $(S_t, A_t) = (S_{t'}, A_{t'})$ for some $t' < t$, continue (first visit MC)
     - $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha_t\big(G_t - Q(S_t, A_t)\big)$ (usually use $\alpha_t = \frac{1}{N(S_t, A_t)}$)

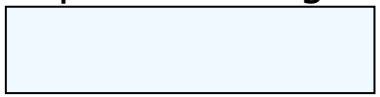
TAQ. Where is policy improvement done?

Off-policy MC control

- Importance sampling

$$\mathbb{E}_{X\sim p(x)}[f(X)] = \mathbb{E}_{X\sim q(x)}\left[f(X)\frac{p(X)}{q(X)}\right]$$

$$\approx \frac{1}{n}\sum_{i=1}^{n} f(x^{(i)})\frac{p(x^{(i)})}{q(x^{(i)})}, \text{ where } x^{(1)},\dots,x^{(n)}\sim q(x)$$

- Off-policy
  - behavior policy ≠ target policy
- Importance sampling
  - behavior policy μ : allowing exploration (such as ε-greedy)
    - $\tau_t = (S_t, A_t, R_t, S_{t+1}, A_{t+1}, R_{t+1}, \dots, S_T)$: sampled from μ
  - target policy π : greedy policy
  - importance weight
    - $P(\tau_t|S_t,\pi) = \prod_{k=t}^{T-1}\pi(A_k|S_k)p(S_{k+1}|S_k,A_k)$
    - $P(\tau_t|S_t,\mu) = \prod_{k=t}^{T-1}\mu(A_k|S_k)p(S_{k+1}|S_k,A_k)$
    - $w^{\pi/\mu} = \frac{P(\tau_t|S_t,\pi)}{P(\tau_t|S_t,\mu)} = \prod_{k=t}^{T-1}\frac{\pi(A_k|S_k)}{\mu(A_k|S_k)} = \frac{\pi(A_t|S_t)}{\mu(A_t|S_t)}\frac{\pi(A_{t+1}|S_{t+1})}{\mu(A_{t+1}|S_{t+1})}\cdots\frac{\pi(A_{T-1}|S_{T-1})}{\mu(A_{T-1}|S_{T-1})}$
  - value / policy update
    - value : $Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \alpha_t\left(\square - Q(S_t,A_t)\right)$
    - policy : $\pi(S_t) \leftarrow \arg\max_a Q(S_t,a)$
  - caveat : $\mathrm{Var}\left[w^{\pi/\mu}G_t\right] \gg \mathrm{Var}[G_t]$
    - importance weight $w^{\pi/\mu}$ can vary too much
    - 

Summary of MC methods

- Advantage of MC methods
  - simple, easy to
  - no need for full models
    - can learn directly from interaction with env.
  - no need to learn about all states
    - do we need to know the value of states that will never be visited?
  - does not use Bellman eq.
    - less harmed by violating
- Disadvantage of MC
  - not applicable to
  - high of return → motivation for TD methods
- For control, sufficient exploration is required
  - exploring starts → impossible in many cases
  - on-policy control with policy
  - off-policy control using sampling
    → impractical because of high